

1.1.1 Definition of a "Cloud"

An aggregate of cloud droplets or ice crystals suspended in the air.

Cloud particles need to exist in a sufficiently large concentration to be visible.

Satellites define clouds based on their optical depth. Uncertain over polar regions (ice reflectivity).

1.1.2 Distribution of Clouds across the Globe

In the global annual average, roughly 70% of Earth's surface is covered with clouds.

The cloud cover is 5% - 15% higher over the oceans than over land.

High cloud amounts at ITCZ (10°N - 15°N), SPCZ southeastward expansion, stratiform clouds over subtropical high pressure systems over cold ocean currents.

1.1.3 Importance of Clouds

① Radiative Impact

- Differing influence on energy budget, depending on solar (shortwave) or terrestrial (longwave)
- Cloud types and altitudes

② Hydrological Impact

- Precipitation from clouds delivers water to the surface as rain or snow
- Wet deposition - removes soluble gases and aerosols
- Medium for aqueous-phase chemical reaction

1.2.1 Macroscopic Cloud Types and Properties

① Cumulus Clouds

- Heap, piles of dense and isolated clouds
- Large vertical extent given by the depth of instability
- Formed due to convection, clear edges as droplets stay within the cloud.

② Stratus Clouds

- Uniform and flat with large horizontal extent
- Large-scale vertical motion in statically stable air (e.g. fronts)
- Little internal vertical motion

③ Cirrus Clouds

- Curly with large horizontal extent
- Ice crystals
- Optically thin as the ice crystals can leave as precipitation, hence the diffuse edges

1.2.2 Ten Cloud Types

- Low-level clouds

層雲 Stratus	}	Usually contains more moisture, seen more over the ocean
層積雲 Stratocumulus		
積雲 Cumulus	}	Higher RH in oceanic boundary layer
雨層雲 Nimbostratus		Precipitating clouds.
積雨雲 Cumulonimbus		Low base height and large vertical extent

① Low-Level Layered Cloud

Vertical extent < 1km

(St) Stratus: purely gray, cannot be distinguished from "high fog".

(Sc) Stratocumulus: gray and white, more round air mass than stratus, less separated than Cu.

② Low-Level Clouds with Vertical Extent

(Cu) Cumulus: local instability

(Cb) Cumulonimbus: fueled by more latent heat released from ice formation

capped by the anvil formed at tropopause (smooth fibrous anvils of Cbs are indicative of ice crystals falling with different speeds)
associated with lightning, hail, thunder

(Ns) Nimbostratus: lifted along warm front

- Mid-level clouds

高積雲 Altocumulus	}	Slow upward lifting over large area
高層雲 Altostratus		

(As) Altostratus: uniform or striated gray/blue

higher cloud base than stratus, smaller optical depth than stratus

(Ac) Altocumulus: More in patches than cumulus.

Virga: fall streak from clouds, evaporates or sublimates before touching the ground.

- High-Level Clouds

卷雲 Cirrus	}	All ice
卷層雲 Cirrostratus		
卷積雲 Cirrocumulus		

(Ci) Cirrus: delicate filaments

(Cs) Cirrostratus: white veil, no individual cloud elements, halo effect 有光暈.

(Cc) Cirrocumulus: granular, intersected by small areas of the blue sky, higher water content due to deep convection, higher optical depth.

Cloud v.s. Fog

- Clouds form due to adiabatic expansion and cooling in rising air parcels
- Fog forms as a result of isobaric mixing and touches the ground

1.2.3 Halos

An optical phenomenon of a bright ring around the sun.

- Light is concentrated at the angle of refraction
- Angle of refraction is 22° , if the hexagonal ice crystals are perpendicular to the light ray and 46° if the hexagonal ice crystals are parallel to the light ray.
(46° halo as outer ring)

1.2.4 Other Cloud Types

Banner Cloud

- Sharp peak splits the air above
- Divergence causes leeward upwelling
- Upwelling air condenses and the cloud formed is advected downward

Roll Cloud

- Formed from thunderstorm cold front or land-sea breeze

Mammatus Cloud

- Pouches underneath the anvil of thunderstorm
 - ▷ Anvil descends, subcloud warms along the dry adiabat, descending cloud momentarily warms along the wet adiabat, cloud warms slower than sub-cloud, differential warming and overturning
 - ▷ High concentrations of heavy hydrometeors, evaporative cooling, overshoots before being restored to neutral buoyancy

Asperatus Cloud

- Roughened wavy cloud base, sinking cloudy air at decaying stage

Kelvin-Helmholtz Wave Cloud

- Two layers of air traveling at different speeds leading to barotropic instability and wave pattern, short-lived

1.3.1 Phases of Clouds

Warm clouds consists purely of liquid water, can be below 0°C (supercooling) as long as ice particles are absent.

Cold clouds can consist purely of ice crystals (ice clouds) or mixture of ice crystals and cloud droplets (mix-phased)

Below -38°C , water freezes homogeneously

1.3.2 Size Distributions

The number size distributions of hydrometeors are described by Gamma distribution

$$n_N(r) = N_0 r^k \exp(-\Delta r)$$

The smallest hydrometeor is a cloud droplet.

$$N_x = \int_0^{\infty} n_N(r) dr$$

$$WC = \int_0^{\infty} m(r) n_N(r) dr$$

Number concentration affects the average size of hydrometeors, precipitation formulation, and albedo

1.3.3 Clouds over Land and Ocean

- Lower heat capacity over land $\rightarrow$ stronger updrafts $\rightarrow S \uparrow \rightarrow$ activation of CCN $\rightarrow$ higher droplet conc.
- Land has more CCN than ocean $\rightarrow$ smaller r , larger N .

Higher heights

- Larger droplets by collision-coalescence growth
- Less water content, less aerosol and lower N .

1.3.4 Polluted Clouds

Higher aerosol concentration, higher cloud droplet concentration at smaller radius

Exercise 1. Solution. $V = 1500 \times 1000 \times 1000 = 1.5 \times 10^9 \text{ (m}^3\text{)}$.

Number of cloud droplets $= 90 \times 10^6 \times 1.5 \times 10^9 = 1.35 \times 10^{17}$

$$m = n \cdot \rho_w \cdot \frac{4}{3} \pi r^3 = 1.35 \times 10^{17} \times 1000 \times \frac{4}{3} \pi (5 \times 10^{-6})^3 \\ = 70686 \text{ (kg)}.$$

Exercise 2. Solution. $pS = mg \quad m = \frac{pS}{g} = \frac{98500 \times 4\pi (6370 \times 10^3)^2}{9.81} = 5.120 \times 10^{18} \text{ (kg)}$

Exercise 3. Solution. (a) $\bar{d} = \frac{0.893}{M_e^{1/3}} \cdot \bar{r}_d = \frac{8.93}{(8.901 \times 10^{-5})^{1/3} \times (5 \times 10^{-6})} = 1.00 \times 10^{-3} \text{ (m)}$

$$M = \frac{4}{3} \pi \bar{r}^3 \cdot \rho_w \cdot N_c \\ = \frac{4}{3} \pi (5 \times 10^{-6})^3 \times 1000 \times 170 \times 10^6 \\ = 8.901 \times 10^{-5} \text{ (kg} \cdot \text{m}^{-3}\text{)}$$

$$(b) \quad d' = \frac{8.93}{M_e'^{1/3}} \cdot \bar{r}_d = \frac{8.93}{(\frac{4}{3} \pi \bar{r}^3 \cdot \rho_w \cdot 3N_c)^{1/3}} \bar{r}_d = \frac{1}{3^{1/3}} d.$$

Exercise 4. Solution. (a) $t = \frac{13000 - 500}{12.5} = 1000 \text{ (s)} \approx 16.7 \text{ min}$

(b) Typical thunderstorm takes 30-60 minutes to develop

Rapid updraft leads to intense convection

(c) Anvil-shape cloud forms. Further vertical ascent is strongly inhibited by the stratospheric layer.

Exercise 5. Solution. (a) $900/0.01 = 90000 \text{ s} = 25 \text{ hr}$ longer lifetime for more aerosol fighting for
oc $500/0.01 = 50000 \text{ s} = 13.89 \text{ hr}$.

(b) Smaller than the computed value, stratus clouds are primarily maintained by large-scale turbulent mixing and condensation rather than steady laminar uplift. Parcel may enter and leave the cloud through mixing without finishing a full base-to-top journey.

(c) Cumuliform clouds are associated with strong updrafts and hence the lifetime strongly agrees with the computed value.

2.1.1 Isobaric Mixing

- Constant pressure, e.g. fog formation, steam over a hot sink
- Energy balance

Heat lost by warmer air = Heat gained by cooler air

$$m_2(C_p + \underbrace{q_{v2}C_{pv}}_{\text{specific humidity}})(T_2 - T_{\text{mix}}) = m_1(C_p + \underbrace{q_{v1}C_{pv}}_{\text{specific humidity}})(T_{\text{mix}} - T_1)$$

Assuming $q_v C_{pv} \ll C_p$.

$$m_2(T_2 - T_{\text{mix}}) = m_1(T_{\text{mix}} - T_1)$$

$$\text{Hence, } T_{\text{mix}} = \frac{m_1 T_1 + m_2 T_2}{m_1 + m_2}, \quad V_{\text{mix}} = V_1 + V_2$$

Heat gained = Latent heat released

$$C_p dT = -L_v dq_v$$

$$q_v = \frac{m_v}{m_m} = \frac{p_v}{p_m} = \frac{p_v}{p_d + p_v} = \frac{e/R_v T}{(p-e)/R_d T + e/R_v T} = \frac{e e}{p - e + e p} \approx e \frac{e}{p}$$

$$e = \frac{M_w}{M_d} = \frac{18}{29} \approx 0.62$$

$$\frac{de}{dT} = \frac{p}{e} \frac{dq_v}{dT} = -\frac{p C_p}{e L_v} \quad \text{isobaric condensation}$$

2.1.2 Adiabatic Mixing

Consider some small-scale turbulent mixing

$$T_{\text{mix}} = \frac{1}{\Delta p} \int_{p_2}^{p_1} \theta dp, \quad q_{v, \text{mix}} = \frac{1}{\Delta p} \int_{p_2}^{p_1} q_v dp$$

Constant across the boundary layer.

2.2 Convection

- Converts potential energy to kinetic energy.
- By having ascending warmer and less dense air parcels being replaced by descending colder and denser air parcels.

• LFC (Level of Free Convection)

The level beyond which an air parcel rises freely due to positive buoyancy. The parcel is warmer than the environment at this threshold.

• LCL (Lifting Condensation Level)

Moist air lifted dry adiabatically saturates.

The dry adiabat intersects with the saturation specific humidity at LCL.

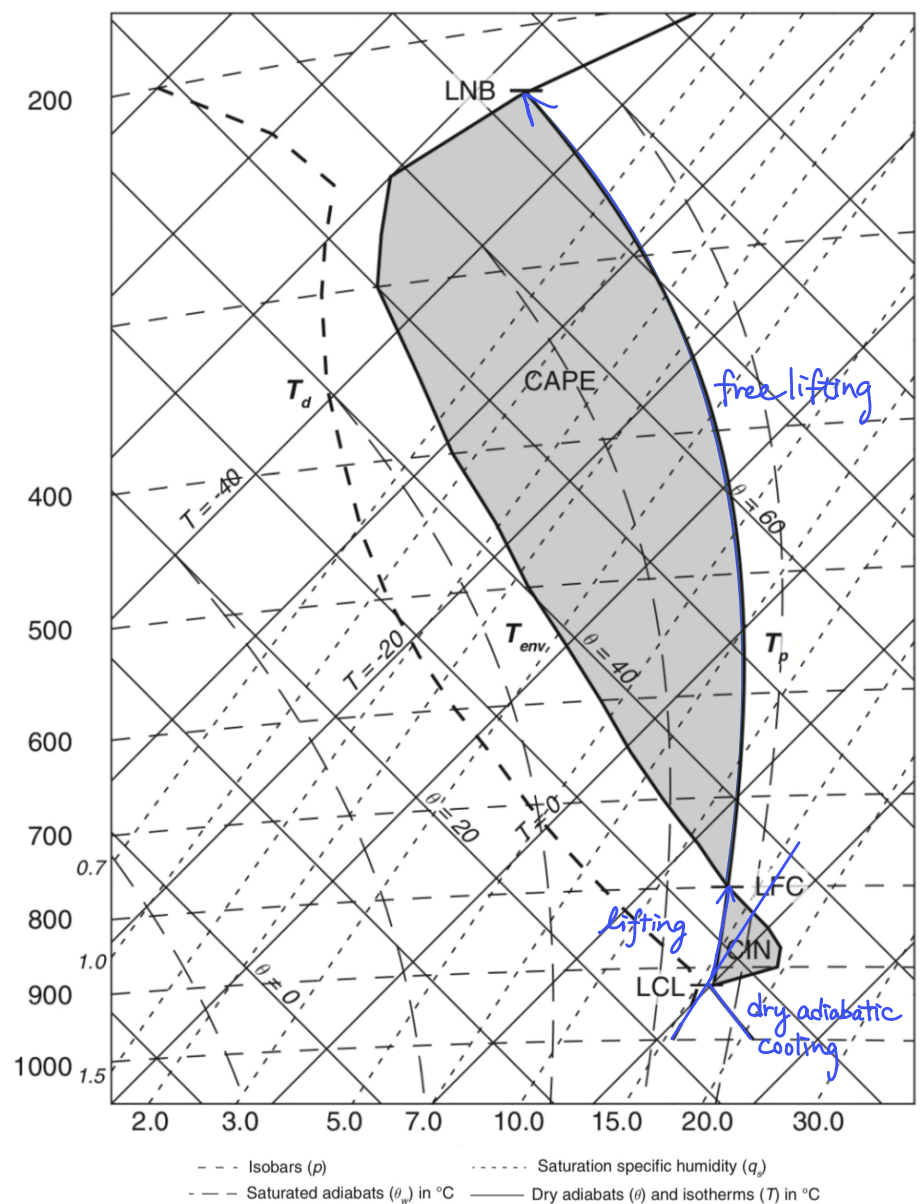
- An air parcel can experience orographic or frontal lifting to reach LFC.

• CCL (Convective Condensation Level)

LCL due to surface heating

• LNB (Level of Neutral Buoyancy)

Threshold of where $T_{\text{env}} > T_{\text{parcel}}$.



- If inversion of temperature appears on the surface, the risen parcel (due to mechanical lifting) can only form stratus but not cumulus; In the morning, heating of the surface allows the LCL to get across the inversion, and the parcel may rise along the saturated adiabat and form cumulus clouds.

2.3 Elementary Parcel Theory

$$F_{\text{upward}} = gV(\rho_2 - \rho_1) = V\rho_1 \frac{d^2(\delta z)}{dt^2}$$

$$\frac{d^2(\delta z)}{dt^2} = g\left(\frac{\rho_2}{\rho_1} - 1\right) = g\left(\frac{T_1}{T_2} - 1\right) \quad \text{Isobaric}$$

$$T_v = T(1 + 0.608 q_v) \quad \text{Correction with water vapor content.}$$

$$\frac{dw}{dt} \cdot dz = g\left(\frac{T_v - T_{v,env}}{T_{v,env}}\right) dz$$

$$w^2 = w_0^2 + 2 \underbrace{\int_{z_0}^z g\left(\frac{T_v - T_{v,env}}{T_{v,env}}\right) dz}_{\text{CAPE}} = w_0^2 + 2 \int_{p_0}^p \frac{R}{p} (T_v - T_{v,env}) dp.$$

$$\text{Hydrostatic balance} \quad dp/dz = -\rho g$$

Convectively available potential energy

Fix CAPE, with lower LNB. Larger $T - T_{v,env} \rightarrow$ larger acceleration.

CIN - convective inhibition

2.4 Modification on the Elementary Parcel Theory

① Environmental Downdraft

$$\rightarrow \text{Air mass exchange} \quad T_{z_0} + (\gamma - I_s) w \delta t < T_{z_0} + (I_d - \gamma) w_{env} \delta t \quad (\gamma - I_s)/(I_d - \gamma) < w_{env}/w.$$

Absolute stability is easier to satisfy, $w_{max} > w_{obs}$.

$$A_w = A_{env} w_{env} \quad \gamma < I_s + (A/A_{env})(I_d - \gamma)$$

② Weight of Condensed Water

$$T_{v,cl} = T(1 + 0.608 q_v - q_c - q_i)$$

③ Dilution by Entrainment

$$S = c_p T + g z + L_v q_v \quad \text{moist static energy} \quad \Delta S = \frac{1}{m} (S_{env} - S_{cl}) \cdot \Delta m$$

$$\frac{dS_{cl}}{dt} = \frac{1}{m} \left(\frac{dm}{dt} \right) \epsilon (S_{env} - S_{cl}) \quad \frac{d}{dt} S = \frac{1}{m} \frac{dm}{dt} [c_p (T_{env} - T_{cl}) + L_v (q_{env} - q_{cl})]$$

④ Aerodynamic Drag

$$\frac{dT_{cl}}{dt} = -\frac{g}{c_p} - \frac{L_v}{c_p} \frac{dq_{cl}}{dt} + \frac{1}{m} \left(\frac{dm}{dt} \right) \epsilon [c_p (T_{env} - T_{cl}) + \frac{L_v}{c_p} (q_{env} - q_{cl})].$$

Friction and drag from environment. (i) dry adiabatic (ii) latent heat (iii) entrainment

$$w_{eff} = (CAPE)^{1/2} \quad \text{instead of } (2 \times CAPE)^{1/2}.$$

Exercise 1. Solution. Fogs are formed by isobaric mixing, and clouds are formed due to adiabatic cooling.

Exercise 2. Solution. $T_{\text{mix}} = \frac{T_1 m_1 + T_2 m_2}{m_1 + m_2} = \frac{2 \times 12 + 1 \times 20}{2 + 1} = 14.67^\circ\text{C}$.

Exercise 3. Solution. $T_{\text{mix}} = \frac{T_{\text{env}} + T_{\text{breath}}}{2} = 18.5^\circ\text{C}$ Clausius-Clayperon $6.109 \exp\left(\frac{17.625T}{243.04 + T}\right) \text{ hPa}$
 $e_s(0^\circ\text{C}) = 6.112 \text{ hPa}$ $e_s(37^\circ\text{C}) = 62.9$ $e_s(18.5^\circ\text{C}) = 21.28 \text{ hPa}$.

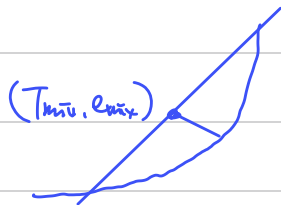
$$e_{\text{env}} = RH_{\text{env}} \cdot e_s = 90\% \times 6.112 = 5.501 \text{ (hPa)}$$

$$e_{\text{breath}} = RH_{\text{breath}} \cdot e_s = 95\% \times 62.9 = 59.755 \text{ (hPa)}$$

$$e_{\text{mix}} = \frac{5.501 + 59.755}{2} = 32.628 \text{ (hPa)} \quad e_{\text{mix}} > e_s(18.5^\circ\text{C}),$$

Yes.

$$\frac{de}{dT} = -\frac{p c_p}{\epsilon L_v} = -\frac{1013 \times 1005}{0.622 \times 2.5 \times 10^6} = -0.655 \text{ hPa/K}.$$



3.1.1 Aerosols

A dispersed system of solid or liquid particles suspended in air
Each cloud droplet forms on an aerosol particle.

3.1.2 Chemical Characteristics of Aerosol Particles

Example: carbonaceous aerosol particles - combustion-generated particles

- Black carbon (BC): Produced by incomplete combustion of fossil fuels and biomass
- Organic carbon (OC): Carbon is chemically bonded to H, O, S, N

Mixing States

- Externally mixed: each particle only contains one species
- Internally mixed: contains different species, due to condensation on a particle, or water evaporating with interlocking species

3.1.3 Physical Characteristics of Aerosol Particles

Aqueous aerosol particles: promotes cloud droplet in warm clouds e.g. sulfuric acid particles

Solid/Crystalline particles: promotes ice crystals in ice & mixed phase clouds, e.g. mineral dust

Nucleation mode	1.5 - 5 nm
Atten mode	5 nm - 0.05 μ m
Accumulation mode	0.05 μ m - 0.5 μ m
Coarse mode	0.5 μ m - 5 μ m
Giant mode	> 5 μ m

Coarse aerosols: typically directly emitted into the atmosphere, e.g. mineral dust, sea salt particles

Nucleation aerosols: formed by supersaturated vapors

3.2.1 Size, Surface, and Volume Distribution of Aerosols

Aerosol number distribution peaks at small radii.

For surface area and volume distribution, usually peaked at large radii.

3.2.2 Observed Aerosol Number Distribution

Terrestrial biomass burning: lowest during wet season: ① less outdoor burning ② washout

Marine aerosols: much lower concentration than land, contributed by anthropogenic sources

Biomass burning in Amazonian forest cause higher conc. of accumulation mode aerosols

Desert dust (coarse mode) over Cape Verde due to desert dust aerosols from Sahara Desert.

3.3.1 Aerosol Sources

Primary particles are emitted directly into the atmosphere, e.g. dust, pollen, combustion, sea salt

Secondary particles are formed within the atmosphere from precursor gases



Sea salt abundance: $40^\circ\text{S} - 50^\circ\text{S}$, ocean with few landmasses and over NH storm tracks

Dust abundance: over Sahara and Gobi deserts

Sulfate emission: power plants and industry, phytoplankton emission of ^{dimethylsulfide} DMS, volcanoes, domestic biofuel, shipping lanes

BC & POM: incomplete combustion of biomass

Nitrate: secondary aerosol forms from gas to particle conversion

3.3.2 Aerosol Sinks

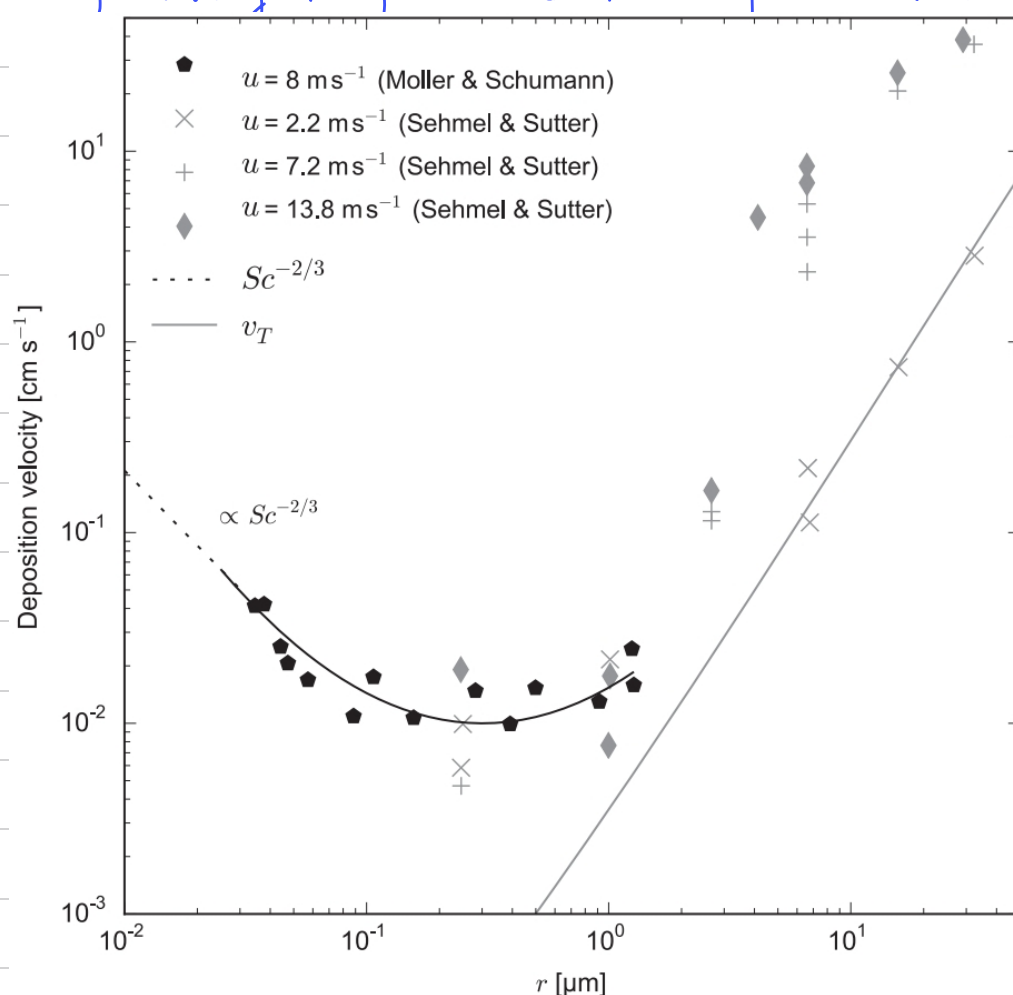
Dry deposition: removal by diffusion & gravity, usually coarse aerosols

Wet deposition: removal by hydrometeors, act as CCN, usually finer molecules

3.4 Dry Scavenging

Smaller particles have a longer diffusion distance (i.e. longer mean free path), touching the surface more easily.

The higher velocity at larger size is due to the higher terminal velocity.



3.5 Wet Scavenging

Dominant sink for Aitken and accumulation mode particles.

① Impaction Scavenging Three major forces: inertial, brownian, phoretic

- Collection efficiency ($\tilde{E}$) = Collision efficiency (E) $\times$ coagulation efficiency ($\hat{E}$)

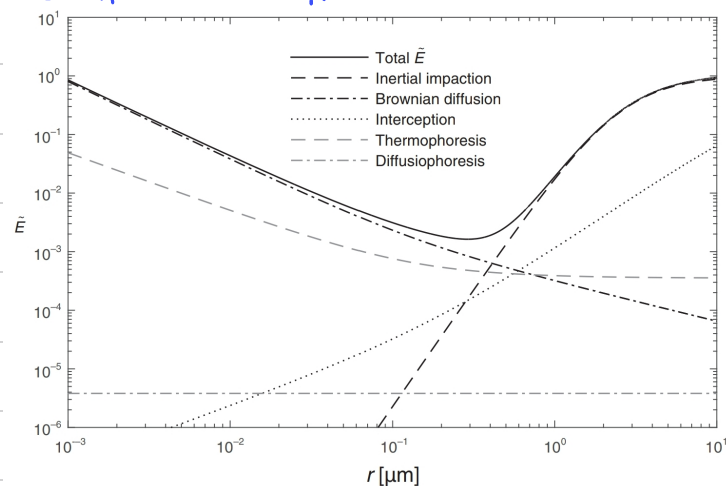
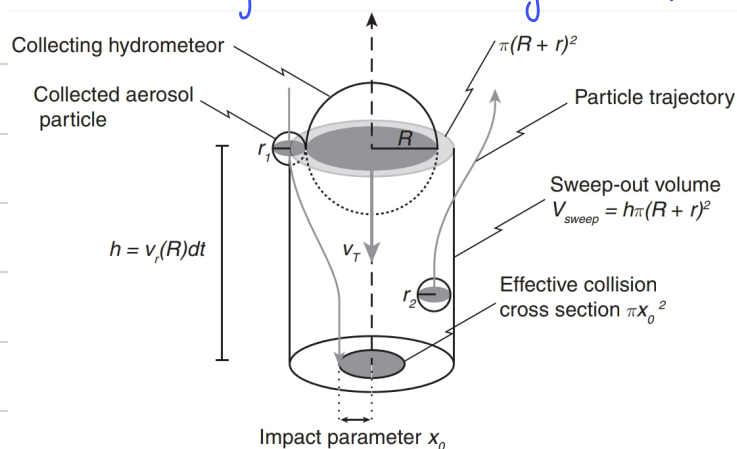
- Takes place in- and below-cloud.

- The area being swept-out is $\pi(R+r)^2$

Impact parameter x_0 partly depends on r . Aerosols further away from x_0 would be deflected

$$E(R, r) = \frac{\pi x_0^2}{\pi(R+r)^2}$$

- larger aerosols $\rightarrow$ larger inertial $\rightarrow$ more difficult to be deflected.



- Phoretic forces

Dominating

Thermophoresis Evaporating hydrometeor attracts neighboring aerosols with higher temperature and kinetic energy (when evaporating)

Diffusiophoresis Higher water vapor density on the hydrometeor's surface drifting aerosols away

② Nucleation Scavenging

- Ability of the aerosol to act as cloud condensation nuclei (CCN)

- Aerosols removed from surrounding air and incorporated into hydrometeor

- Takes place in-cloud only

- Best at removing accumulation size aerosols, lower supersaturation to activate as CCN.

3.6 Aerosol Abundance and Lifetime

Difference between formation and removal contributes to aerosol abundance

- Dust: coarse mode/accumulation mode, can be transported upward by dust storms or deep convection, hence the longer lifetime.

- Sea salt: mainly in coarse modes, remains near the surface, can be quickly deposited away from the atmosphere

$$\text{Lifetime } (\tau) = \frac{\text{Abundance}}{\text{Flux}}$$